

(12) **United States Patent**
Mukherjee et al.

(10) **Patent No.:** **US 12,402,006 B1**
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **SYSTEMS AND METHODS FOR IDENTIFYING TRANSMITTERS**

(71) Applicants: **Tathagata Mukherjee**, Huntsville, AL (US); **Eduardo Pasillao**, Fort Walton Beach, FL (US); **Debashri Roy**, Fort Walton Beach, FL (US); **Mainak Chatterjee**, Orlando, FL (US); **Erik Blasch**, Dayton, OH (US)

(72) Inventors: **Tathagata Mukherjee**, Huntsville, AL (US); **Eduardo Pasillao**, Fort Walton Beach, FL (US); **Debashri Roy**, Fort Walton Beach, FL (US); **Mainak Chatterjee**, Orlando, FL (US); **Erik Blasch**, Dayton, OH (US)

(73) Assignees: **Board of Trustees of The University of Alabama, for and on Behalf of The University of Alabama in Huntsville**, Huntsville, AL (US); **University of Central Florida**, Orlando, FL (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 1023 days.

(21) Appl. No.: **17/211,435**

(22) Filed: **Mar. 24, 2021**

Related U.S. Application Data

(60) Provisional application No. 63/008,556, filed on Apr. 10, 2020, provisional application No. 62/993,751, filed on Mar. 24, 2020, provisional application No. 62/993,748, filed on Mar. 24, 2020.

(51) **Int. Cl.**
H04L 9/32 (2006.01)
H04L 9/40 (2022.01)
H04W 12/122 (2021.01)

(52) **U.S. Cl.**
CPC **H04W 12/122** (2021.01); **H04L 63/1483** (2013.01)

(58) **Field of Classification Search**
CPC H04W 12/122; H04L 63/1483
USPC 713/176
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2015/0077129 A1* 3/2015 Zinevich G01S 5/06
324/512
2020/0213354 A1* 7/2020 Chakraborty G06F 9/4416
2020/0398083 A1* 12/2020 Adelsheim G06N 3/045

* cited by examiner

Primary Examiner — Jeffrey C Pwu

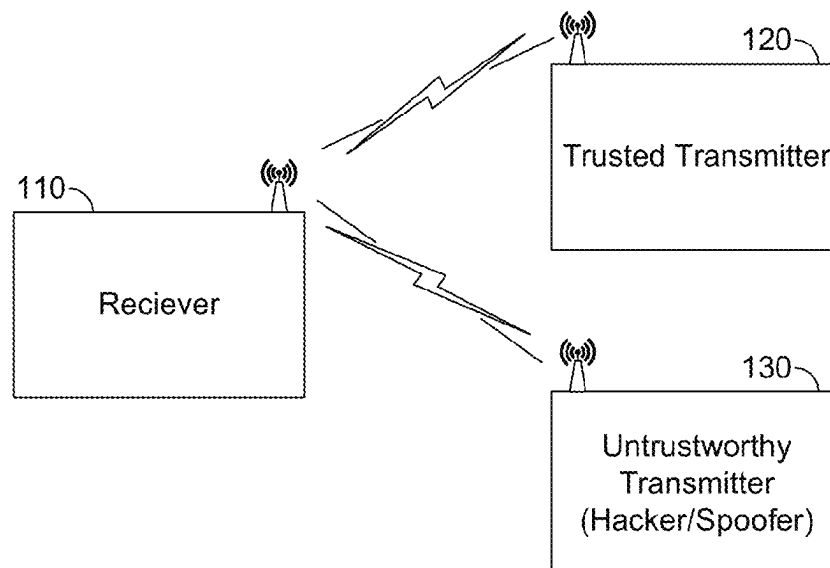
Assistant Examiner — Samuel Ambaye

(74) *Attorney, Agent, or Firm* — Jon E. Holland; Butler Snow LLP

(57) **ABSTRACT**

A signal transmitted by a trusted transmitter is sampled, and a machine-learning algorithm is trained using these samples to learn to distinguish the signal characteristics of the trusted transmitter from the signal characteristics of other transmitters. The machine-learning parameters learned to distinguish the trusted transmitter's signals can then be exported allowing other devices to authenticate signals from the trusted transmitter. Since the signal characteristics of the trusted transmitter is based on manufacturing variations, the transmitters of unauthorized users should be unable to replicate the signal characteristics of the trusted transmitter. Thus, the transmitters of unauthorized users should not be usable to spoof a device that uses the exported parameters to authenticate messages from the trusted transmitter.

12 Claims, 9 Drawing Sheets



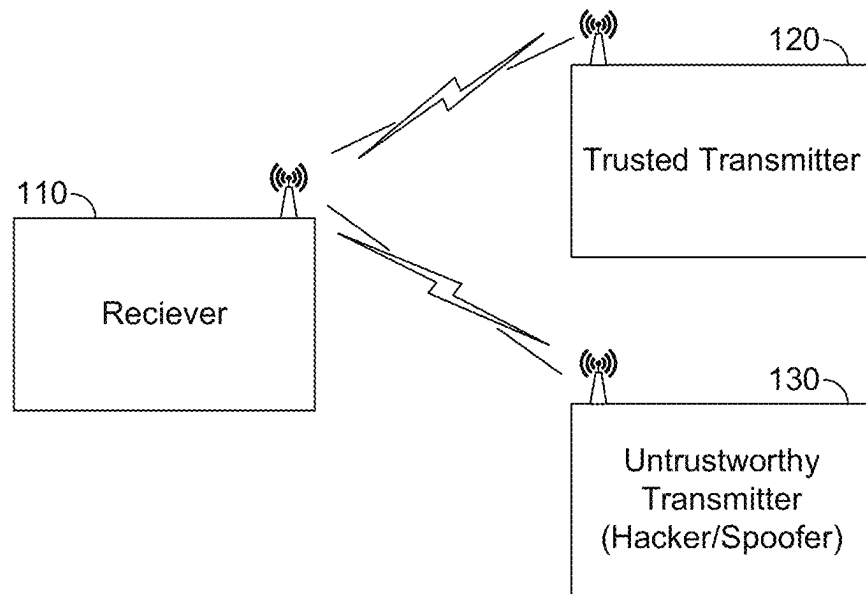
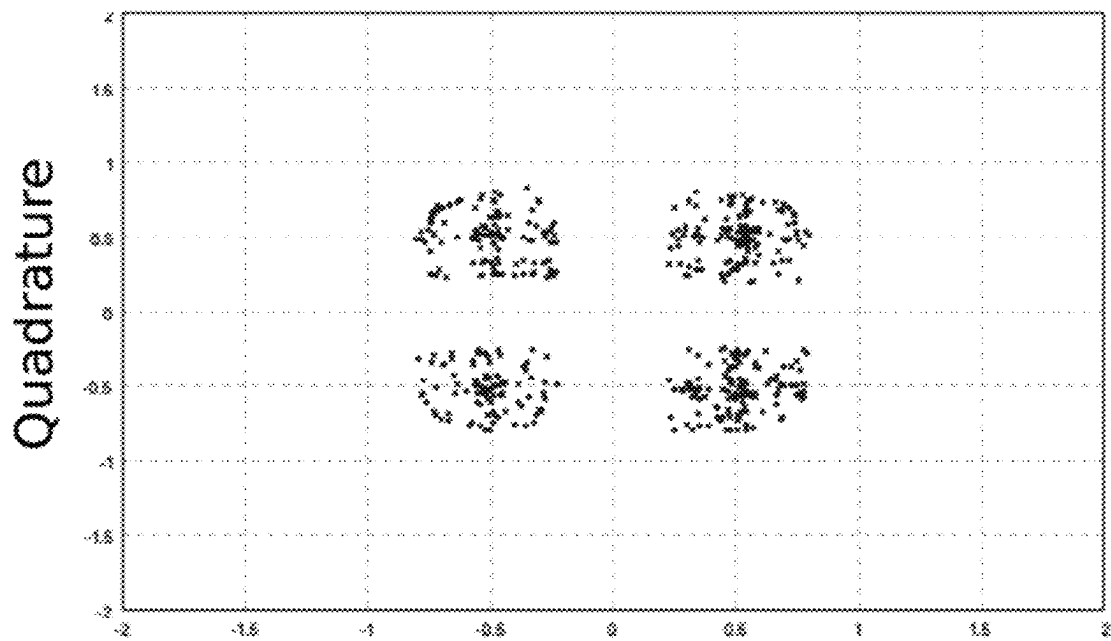
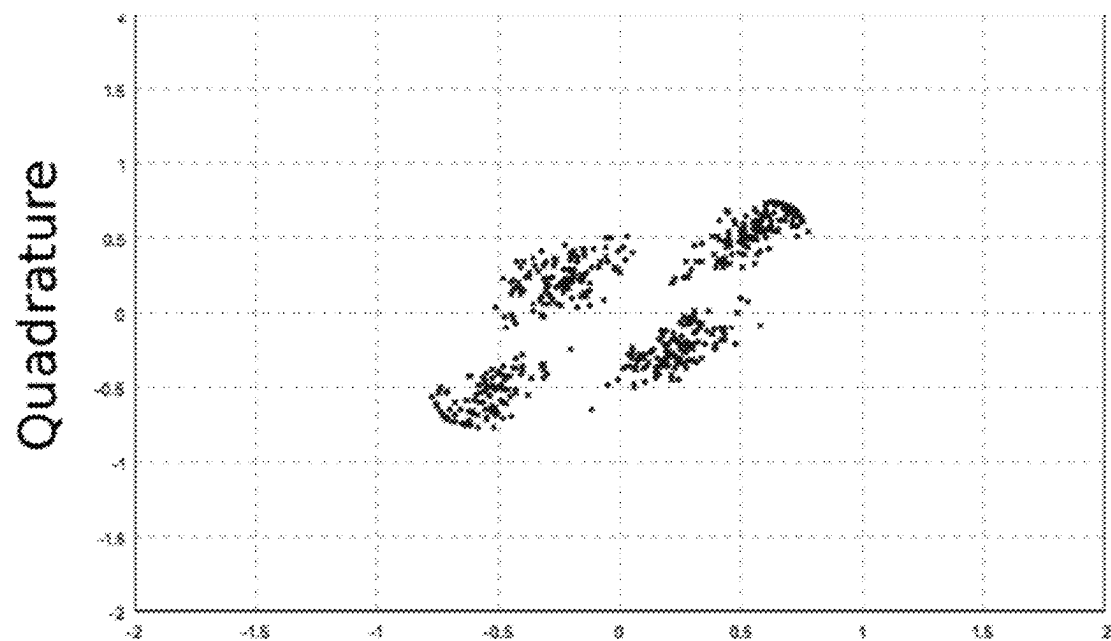


FIG. 1



Inphase

FIG. 2a



Inphase

FIG. 2b

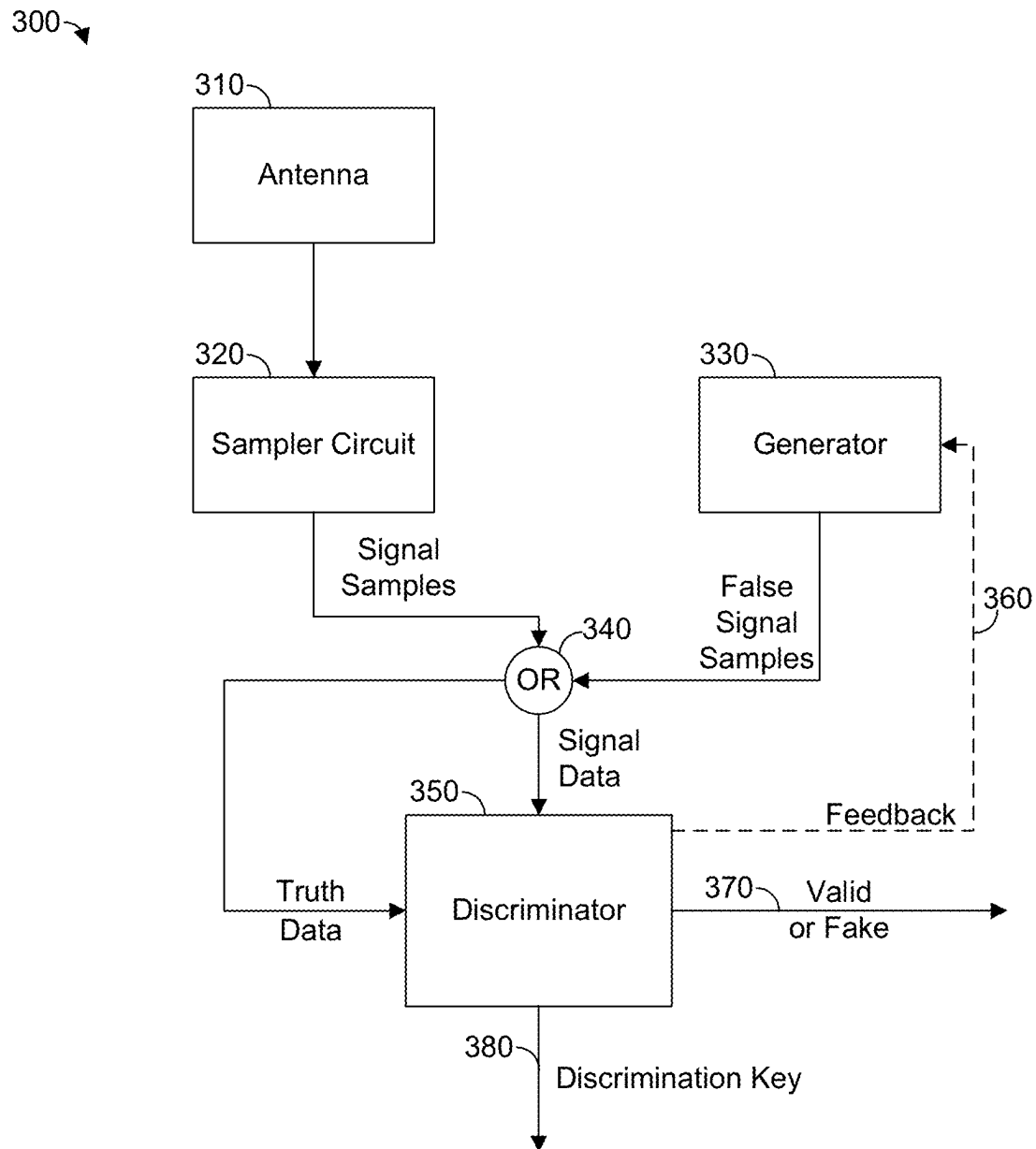


FIG. 3

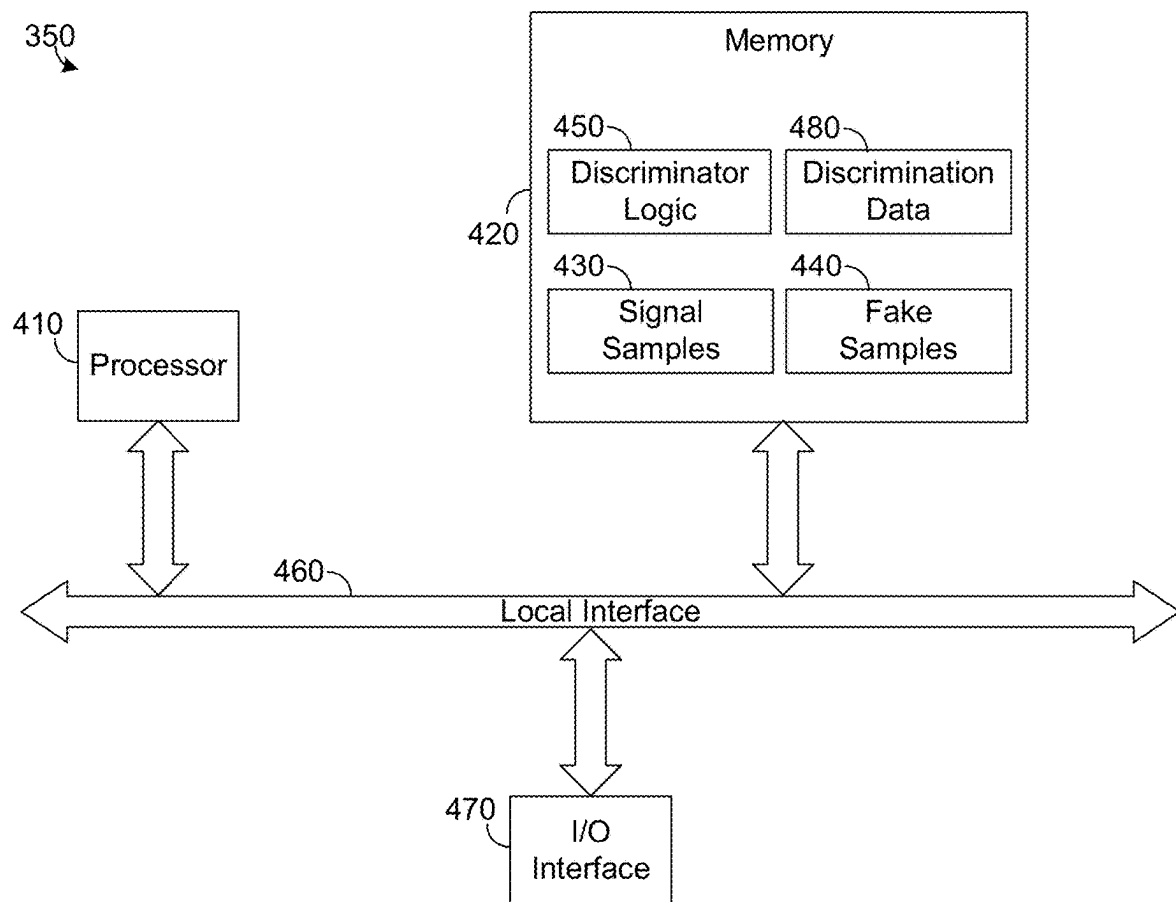


FIG. 4

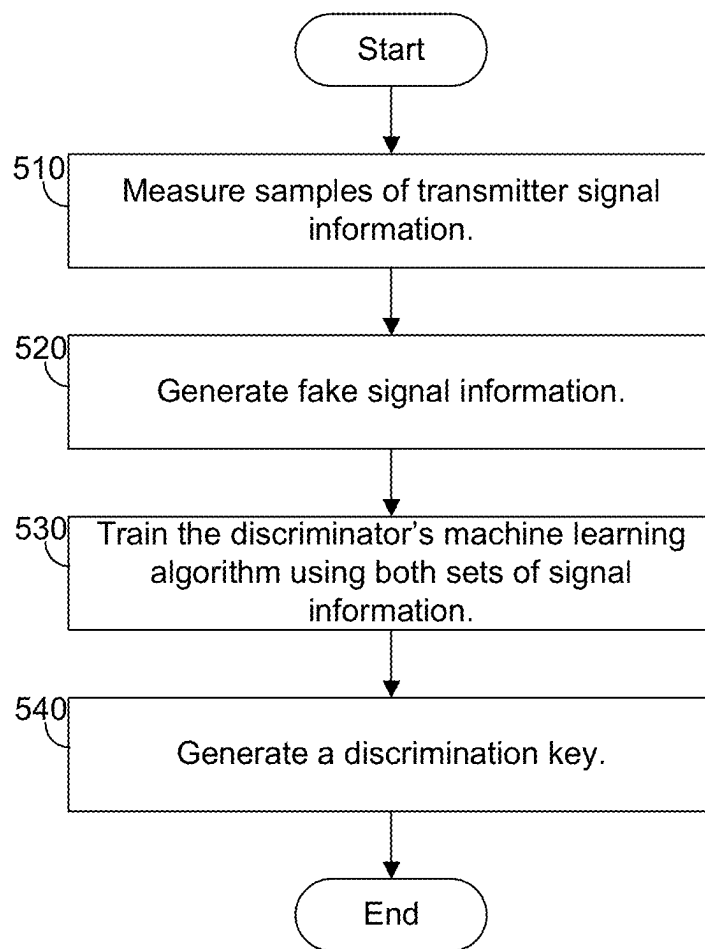


FIG. 5

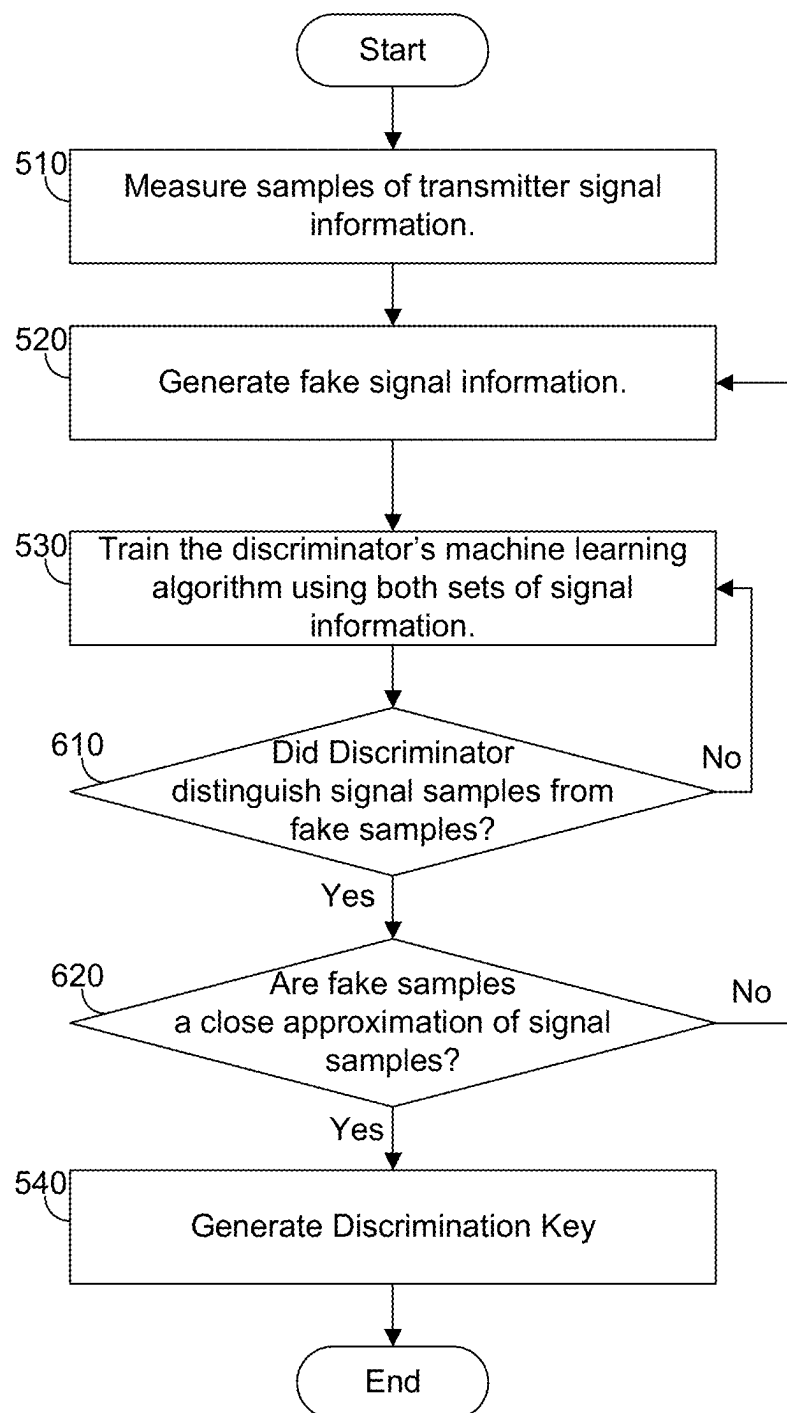
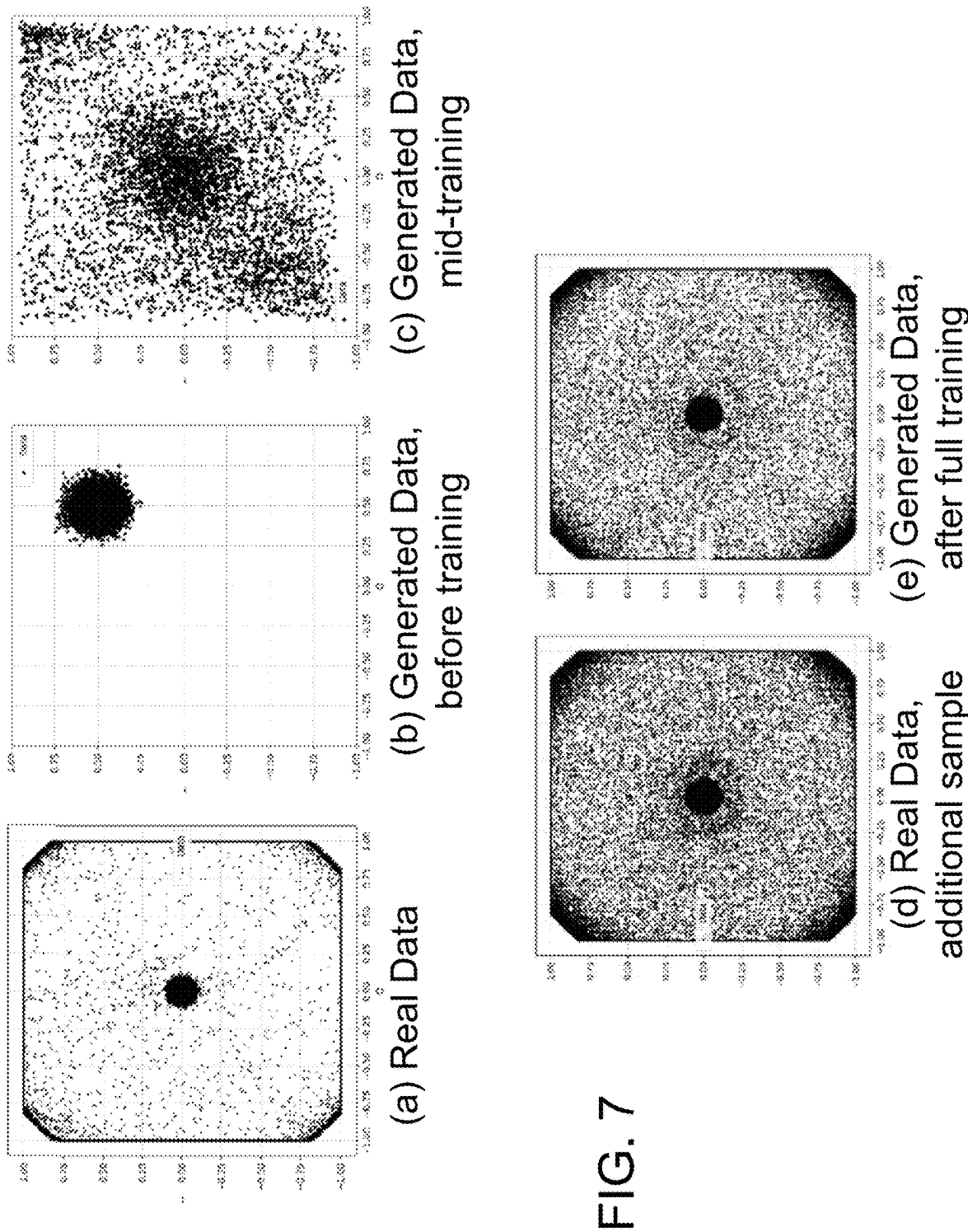


FIG. 6



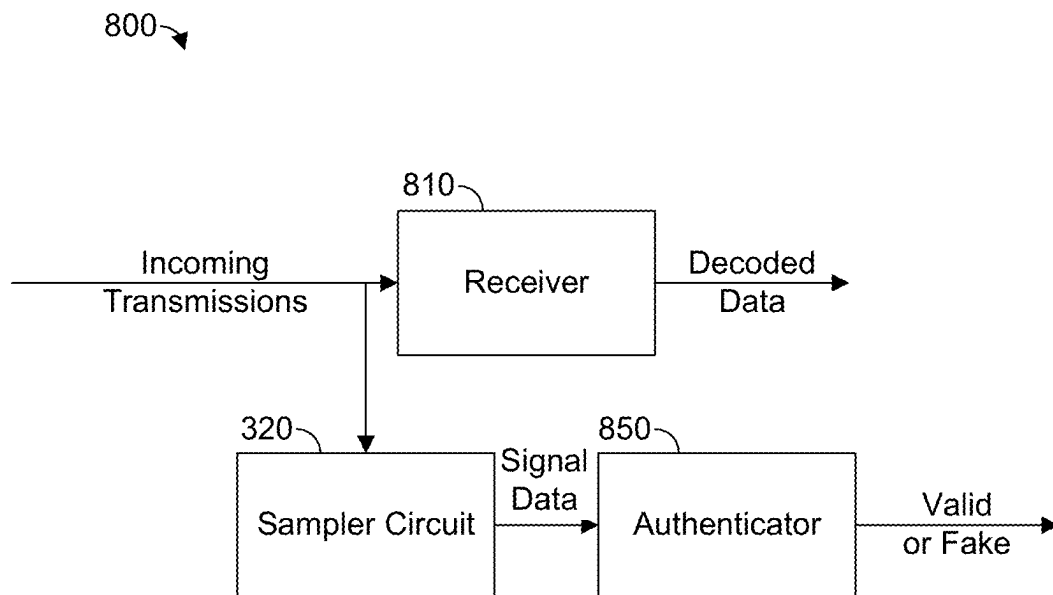


FIG. 8a

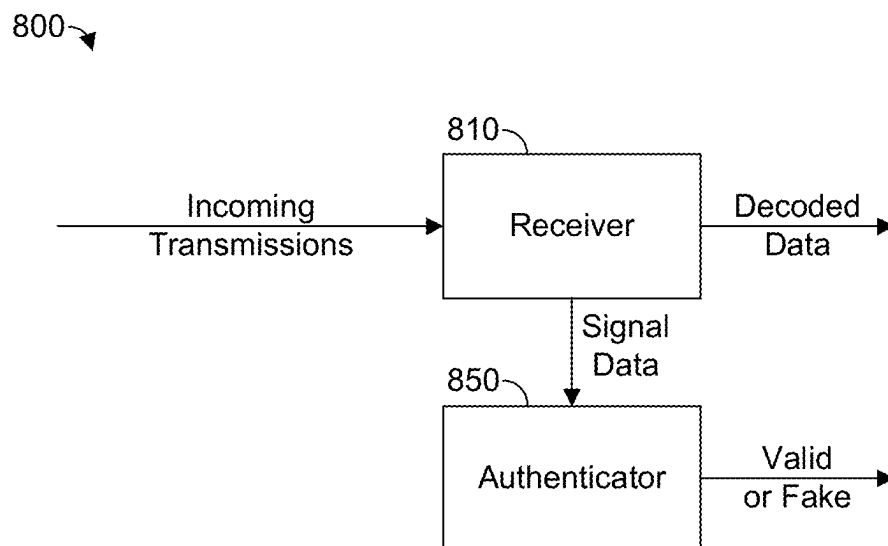


FIG. 8b

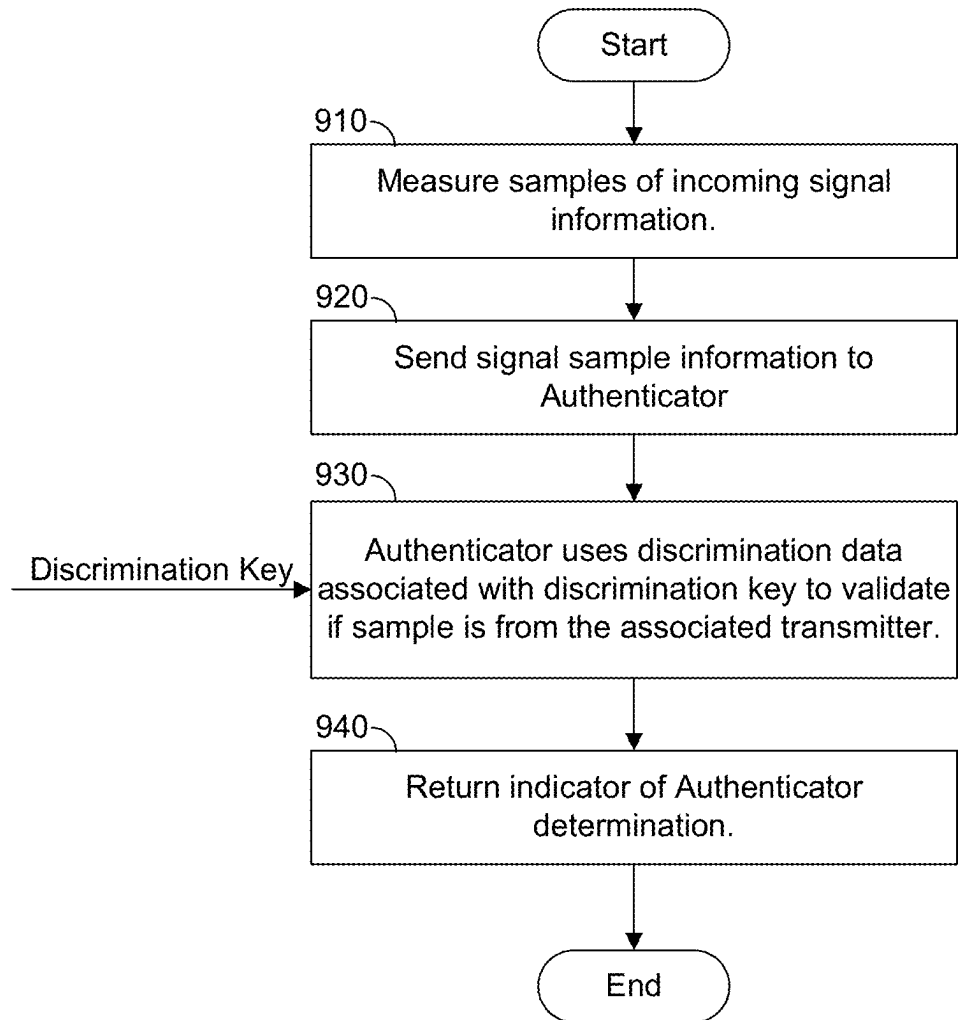


FIG. 9

1

SYSTEMS AND METHODS FOR IDENTIFYING TRANSMITTERS**CROSS REFERENCE TO RELATED APPLICATIONS**

This application claims priority to U.S. Provisional Application No. 62/993,748, entitled “Transmitter Identification Using Adversarial Networks and In-Phase and Quadrature (IQ) Data” and filed on Mar. 24, 2020, which is incorporated herein by reference. This application also claims priority to U.S. Provisional Application No. 62/993,751, entitled “Radio Frequency Adversarial Learning (RFAL) for RF Transmitter Identification and Classification” and filed on Mar. 24, 2020, which is incorporated herein by reference. This application further claims priority to U.S. Provisional Application No. 63/008,556, entitled “Systems and Methods for Identifying Transmitters” and filed on Apr. 10, 2020, which is incorporated herein by reference.

FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

This invention was made with Government support under contracts Air Force Research Laboratory Contract FA8651-16-2-0009. The Government has certain rights in the invention.

RELATED ART

Wireless communication has been a popular form of communication for many years. Indeed, wireless communication is often used to make financial and other types of transactions involving sensitive information, such as user names, passwords, account numbers, or other information that users wish to keep private.

An important aspect of such communication of sensitive information is authentication of received messages or, in other words, ensuring that received messages have originated from a trusted source. Typically, networks use media access control (MAC) addresses, international mobile equipment identity (IMEI) numbers, and other data in a received message to identify participants (e.g., transmitters) on such networks. Many of these devices and communication protocols are susceptible to a number of attacks and exploits including man-in-the-middle attacks that allow the theft of sensitive data or to insert command and instructions while impersonating another device. Using such exploits, information used to authenticate a source may be cloned, copied, or spoofed by other rogue transmitters for malicious or nefarious purposes. By using such information, malicious actors can use rogue transmitters to impersonate other transmitters allowing the malicious actors to defeat many security protocols and measures.

Various forms of encryption and other verification methods have been used in an attempt to overcome such problems, but they are susceptible to various forms of exploits and impersonation methods. In general, improved techniques for transmitter identification are generally desirable to make it more difficult for unauthorized users or hackers to impersonate a trusted transmitter or otherwise defeat security measures that rely on authentication of received messages.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure can be better understood with reference to the following drawings. The elements of the drawings are

2

not necessarily to scale relative to each other, emphasis instead being placed upon clearly illustrating the principles of the disclosure. Furthermore, like reference numerals designate corresponding parts throughout the several views.

FIG. 1 is a block diagram illustrating an exemplary communication system having a receiver that receives communications from trustworthy and untrustworthy sources.

FIG. 2a is a graph illustrating exemplary plots of a quadrature component of a signal transmitted by a first transmitter versus an in-phase component of the signal.

FIG. 2b is a graph illustrating exemplary plots of a quadrature component of a signal transmitted by a second transmitter versus an in-phase component of the signal.

FIG. 3 is a block diagram illustrating an exemplary system for generating discrimination data.

FIG. 4 is a block diagram illustrating an exemplary discriminator.

FIG. 5 is a flow chart illustrating an exemplary method of generating a discrimination key using machine learning.

FIG. 6 is a block diagram illustrating an exemplary iterative method of generating a discrimination key using machine learning.

FIG. 7a-e illustrate the differences between real signal samples and generated signal samples as the generator is trained to better approximate signal samples.

FIG. 8a is a block diagram illustrating an exemplary communication device configured to authenticate a signal from a trusted transmitter.

FIG. 8b is a block diagram illustrating an exemplary communication device configured to authenticate a signal from a trusted transmitter.

FIG. 9 is a block diagram illustrating an exemplary method of authenticating signals with a discriminator key.

DETAILED DESCRIPTION

The present disclosure generally pertains to systems and methods for identifying transmitters based on signal characteristics. Manufacturing variations of transmitters cause detectable changes in signal characteristics of the transmitted signals. These manufacturing variations result in a signal pattern that is unique to each transmitter even for those manufactured at the same factory. In some embodiments, a signal by a trusted transmitter is sampled, and a machine-learning algorithm is trained using these samples to learn to distinguish the trusted transmitter's signal from the signals transmitted by other transmitters. In some embodiments, the training of the machine-learning algorithm is performed against signals from transmitters attempting to mimic the trusted transmitter's signals. The machine-learning parameters learned to distinguish the trusted transmitter's signals can then be exported allowing other devices to authenticate signals from the trusted transmitter. Since the transmitters of unauthorized users should be unable to replicate the signal characteristics of the trusted transmitter, such transmitters should not be usable to spoof a device that has access to the exported parameters and uses such parameters to authenticate messages from the trusted transmitter.

FIG. 1 is a block diagram illustrating an exemplary receiver 110 receiving communications from a trusted transmitter 120 and an untrustworthy transmitter 130. There are many scenarios involving wireless communications between devices (e.g., cellphones and cell phone towers, laptops and wireless routers/hot spots, key fobs and vehicles). While these systems often provide great benefits to their users, they expose users to many vulnerabilities by which malicious individuals (e.g., thieves, spies, hackers, etc.) can exploit

these systems for nefarious purposes (e.g., steal data, intercept text messages, calls and other communications, unlock cars, start cars, or otherwise impersonate an authorized user). Many of these exploits involve using one transmitter 130 to impersonate a trusted transmitter 120. Such impersonations may include copying and retransmitting data from the trusted transmitter's signal, boosting signal of car key fobs and credit cards to facilitate unauthorized access and activity, man in the middle attacks, using authorization information of the trusted transmitter 120 (e.g., IMEI number, SIM number, cookie information, etc.), and so forth. This can also apply to military communication technologies. By providing a receiver the ability to distinguish one transmitter from another, greater security may be achieved and many of these exploits may be eliminated. For example, if an untrustworthy transmitter attempts to impersonate a trusted transmitter using the proper authentication information, successful impersonation should be preventable if the receiver can distinguish the transmitter based on the incoming signal characteristics. That is, since the characteristics of a transmitted signal are based on manufacturing variations, the untrustworthy transmitter should be unable to replicate the signal characteristics of the trusted transmitter.

There are many aspects of signal characteristics that may form the basis of signal identification and classification (e.g., transient analysis, bi-spectrum matrices, in-phase and quadrature (I/Q) imbalance analysis, etc.). For example, FIGS. 2a and 2b illustrate exemplary in-phase and quadrature differences between different transmitters. In FIG. 2a, a transmitter's broadcast may provide I/Q values that fall generally in four circular- or square-like arrangements. In FIG. 2b, a different transmitter while broadcasting similar or the same data has I/Q values that exhibit an angled compression. That is, the I/Q values of the transmitter for FIG. 2a, representing the signal pattern for a signal transmitted by such transmitter, form a signature that is unique or, in other words, different than the signature of I/Q values for the transmitter for FIG. 2b, representing the signal pattern for a signal transmitted by such transmitter. A system can be configured to identify such differences in a way to identify or distinguish between the sources of a signal by recognizing differences in signal characteristic patterns.

By training machine-learning algorithms on signal samples of a particular transmitter, referred to herein for illustrative purposes as the "trusted transmitter," compared to other signal samples from other transmitters, machine-learning algorithms may learn to distinguish the particular transmitter's signals from other signals. By introducing other signal samples, including signal samples that are very similar to the transmitter's signal samples, the machine-learning algorithms can learn to distinguish the characteristics of the trusted transmitter's signals from the characteristics of even very similar other signals. In some embodiments, the other signal sample set may include samples that are very close to the trusted transmitter's signal samples. In some embodiments, the other signals can be changed over time to be closer to the signal samples from the trusted transmitter, as will be discussed in more detail below. The process of training using similar samples is referred to "adversarial learning." The process of training using increasingly similar other samples generated with feedback from a discriminator is referred to as "generative adversarial learning."

FIG. 3 is a block diagram illustrating an exemplary embodiment of a discrimination key generator (DKG) 300. The DKG 300 receives a signal from a transmitter 110 through one or more antennas 310. The signal is sampled by

a sampler circuit 320. In some embodiments, the sampler circuit 320 may include a digital radio or a software-defined radio (SDR). A generator 330 creates fake signal data to be used in training the discriminator 350. A sample selector 340 selects between signal samples 430 (FIG. 4) and fake signal samples 440 (FIG. 4) to send to the discriminator 350. The discriminator 350 implements a machine-learning algorithm that trains on the signals provided by the sample selector 340. The discriminator 350 also receives truth data from the sample selector 340 to update the discriminator's 350 learning functions. The discriminator 350 outputs a valid or fake determination 370. The discriminator 350 also creates a discrimination key 380 that may be used by other devices to perform the same determination calculated by the discriminator 350 without the need to sample the transmitter signal first or perform any training. In some embodiments, the discriminator 350 may provide feedback 360 to the generator 330 to improve the false signal sample generation to allow the generator 330 to produce fake signals 440 that are closer to the signal samples 430.

The machine-learning algorithm may include any known machine-learning methods including but not limited to regression, classification, clustering, dimensionality reduction, ensemble methods, neural nets and deep learning, transfer learning, reinforcement learning, etc. In a preferred embodiment, signal samples 430 (FIG. 4) include in-phase (I) and quadrature (Q) data and the discriminator 350 uses a form of recurrent neural network (RNN) to discriminate between real and fake transmitter signal samples. In some embodiments, the signal samples may be collected and transmitted via a network to a discriminator 350. Other machine learning techniques may be used. For example, in place of a RNN, a convolutional neural network (CNN) or a fully connected deep neural network (DNN) may be used.

The discriminator 350 may be implemented in hardware (e.g., an application-specific integrated circuit (ASIC), a field programmable gate array (FPGA)), software and/or firmware in conjunction within one or more processors to execute the software and/or firmware, or any combination of hardware, software, or firmware. FIG. 4 illustrates an exemplary discriminator 350. The discriminator 350 may use one or more processors 410 (e.g., digital signal processor or central processing unit), a memory 420, and one or more input output (I/O) interfaces 470 connected through a local interface 460 (e.g., a system bus). The memory 420 may store discriminator logic 450 (including machine-learning functions) and discrimination data 480 (e.g., machine-learning variables, weights, or other parameters or data structures). The discrimination logic 450 may receive signal samples 430, fake samples 440, and truth data (distinguishing the signal samples 450 and the fake samples 440) to train the underlying machine-learning functions. Depending on the underlying machine-learning functions, the training may be iterative or performed all at once. Training of machine-learning functions often involve the setting and adjusting of collections of variables, weights, and other parameters and data structures collectively described here as discrimination data 480. As an example, as described in more detail herein, the discrimination data 480 may define the latent space representation learned by the discrimination logic 450 for encoding the variations in noise of the signal samples 430, noting that such variations are unique to the transmitter 110 being characterized. Thus, the discrimination data 480 represents the noise signature of such transmitter 110 and may be used by other receivers to identify the transmitter 110, as further described below.

After discriminator training has been completed, the discrimination data **480** may be output to allow other devices to distinguish between the signals from the transmitter **110** and other transmitters. Further details about discriminator training will be discussed later. A discrimination key **380** may refer to a serialization of discrimination data **480**, a compression of the discrimination data **480**, or an identifier allowing the query or lookup of the discrimination data **480** from a database or other storage location for the discrimination data **480**.

A discriminator **350** may share hardware resources with other components of the DKG **300**, including one or more processors **410** and memory **420**. As an example, the generator **330** and/or sample selector **340** may utilize the same processors and/or memory as the discriminator **420**. In some embodiments, the I/O interface **470** may be configured to receive signal samples **430**, fake samples **440**, and truth data for use by the discriminator logic **450**.

In some embodiments, the inherent variation in signal generation due to manufacturing variations in transmitters is exploited to allow the identification or classification of the signal sources through analysis of the signals themselves. The DKG **300** receives a signal from a transmitter **120** (e.g., through an antenna **310**). There may be decoders or demodulators to separate one signal from various other signals as known in the art. The signal is passed to a signal sampler **320** that creates signal samples **430** from the received signal. A generator **330** may be used to create fake signal samples **440**. The signal samples **430** and fake signal samples **440** are fed into a discriminator **350**. In some embodiments, the various samples are sent to the discriminator **350** in a random or iterative manner by a signal selector **340** in which case indicators (e.g., truth data) are also sent indicating whether each sample is an actual sample from the signal by the transmitter or a fake sample. The discriminator **350** uses a machine-learning algorithm to learn to distinguish between the signal samples **430** and fake signal samples **440**. The machine-learning variables and parameters developed from such training are used to form the discrimination key **380**.

In learning to identify unique or rare items among a large number of potentially similar and distinct objects, it may be helpful to begin by learning to distinguish between distinct objects first as a simpler case. To become an expert at distinguishing and identifying objects, it can be helpful to learn to distinguish between similar objects. Hence, in some embodiments, the generator **330** may use signal samples **430** or feedback from the discriminator **350** in generating fake signal samples **440**. In some embodiments, during initial training, the fake signal samples **440** might not significantly resemble signal samples **430** during initial machine-learning training. As the discriminator **350** becomes better at distinguishing between signal samples **430** and fake signal samples **440**, feedback from the discriminator **350** may be given to the generator **330** allowing the generator **330** to improve the quality of the fake signal samples **440** or, in other words, to make the fake signal samples **440** to more closely resemble the actual signal samples **430**. Many different machine-learning techniques may be effectively used to perform the classification and identification of signals. Some experiments have shown that, among convolutional neural network (CNN), a fully connected deep neural network (DNN), and a recurrent neural network (RNN), a DNN appears more accurate than a CNN and a RNN is more accurate than a DNN. Experimental data indicates that a RNN with both long short-term memory (LSTM) and gated recurrent units (GRU) may be used to exploit the time series properties of the I/Q data for more consistent and accurate

results. By taking into account the time series in the discriminator analysis, factors that change over time may be discovered by the machine-learning algorithms (e.g., transmitter parts properties change as they heat up changing differently due to slight manufacturing variations).

In some embodiments, the process of improving the fake signal samples **440** and improving the discriminator **330** may be repeated a set number of times, until a threshold quality is achieved, or some combination thereof. In some embodiments, the signal may be in the form of quadrature amplitude modulation (QAM), and signal samples **430** may be in the form of in-phase and quadrature data. In other embodiments, other types of signals and signal data may be used.

FIG. 5 illustrates an exemplary embodiment of a process for generating a discrimination key using machine-learning. Initially, at step **510**, the DKG **300**, measures samples of the trusted transmitter's signal information. For example, a sampling circuit **320** attached to an antenna **310** can sample the trusted transmitter's signal. This sampling may take place, for example, by the transmitter's manufacturer or other user. At step **520**, the generator **330** generates fake signal samples **440**. The fake signal samples **440** can come in the form of purely computer-generated signals, collections of signals from other transmitters (e.g., recordings), or combinations thereof. At step **530**, signal samples **430** and fake signal samples **440** are sent to the discriminator **350** to train the discriminator's machine-learning algorithm. The methods of training are described herein in more detail.

At step **540**, a discrimination key **380** is generated for use in discriminating between signal samples **430** and fake signal samples **440**. This discrimination key **380** can be stored for future use or transmitted to other devices, which may use the discrimination key **380** to authenticate signals, as will be described in more detail below.

As discussed earlier, in some embodiments, the discriminator **350** may be trained by progressively better fake signal samples **440**. Hence, in some embodiments, the discriminator **350** may be trained through an iterative method. The generator **330** may also receive feedback from the discriminator **350**, access to the signal samples **430**, or a combination thereof to improve the quality of the fake signals samples **440** (e.g., to make the fake signal samples **440** a closer approximation to the real signal samples **430**), thereby improving the training of the discriminator **350**. FIG. 6 is a block diagram illustrating an iterative method of generating a discrimination key **380** using machine-learning. Steps **510**, **520**, **530**, and **540** are similar to what has been described previously. At step **610**, the discriminator **350** can determine if the machine-learning functions have properly distinguished signal samples **430** from fake samples **440**. In some embodiments, the discriminator **350** may use a threshold value to determine if the machine-learning functions are sufficiently accurate in distinguishing signal samples **430** from fake samples **440**. If the discriminator **350** is able to distinguish signal samples **430** from fake samples **440** within the threshold accuracy, control passes to step **620**, otherwise control passes to step **530** for further training.

At step **620**, the discriminator **350** determines if the fake samples **440** are a close approximation of the signal samples **430**. If so, control passes to step **540**. Otherwise feedback from the discriminator **350** is returned to the generator **330**, and control returns to step **530** to generate fake samples **440** that are closer approximations of the signal samples **430**. The process of improving the fake signals and retraining the discriminator **350** may repeat successively until the fake samples **440** are determined to be within a threshold varia-

tion of the signal samples **430**. In other embodiments, the process may repeat until the changes made to the fake samples **440** or discrimination machine-learning parameters falls below a predetermined threshold. In yet other embodiments, the process is repeated a specified number of times, or training of the discriminator steps **530** and **610** may be repeated until the accuracy achieved substantially stops improving.

To help illustrate how the fake signal samples may change, FIGS. **7a-e** illustrate the differences between real signal samples **430** and fake samples **440** as the generator **330** is trained to better approximate signal samples **430**. FIGS. **7a** and **7d** illustrate IQ data from signal samples **430** of an exemplary transmitter. FIGS. **7b**, **7c**, and **7e** illustrate examples of fake samples **440** IQ data before training, after partially training, and after full training of the generator **330** respectively. Initially, the differences between the signal samples **430** (FIGS. **7a** and **7d**) and fake sample **440** (FIG. **7b**) may be significant making distinguishing the signal sample **430** and fake sample **440** easy for the discriminator **350**. As shown in FIGS. **7c** and **7e**, as the generator **330** becomes better trained, the differences between signal samples **430** (FIG. **7d**) and fake samples **440** (FIG. **7e**) becomes less apparent. A better trained generator **330** allows for the creation of better fake signal samples **440**. By training on successively better data, the discriminator's machine-learning functions become significantly more adept on distinguishing signal samples **430** from the trusted transmitter than signals from other devices attempting to impersonate the trusted transmitter. Signal patterns that are too similar to be distinguished by humans can be distinguished by the machine-learning algorithms of the discriminator **350**. Once the training has been completed, a discrimination key **380** may be generated at step **540**.

After the discrimination key **380** is defined, the key **380** may be sent to other devices that can use the discrimination key **380** to authenticate signals from the trusted transceiver. That is, another device may use the key **380** to determine that a received signal was in fact transmitted by the trusted transmitter. In this regard, such other device may have an authenticator **850** (FIG. **8**) that receives the discrimination key **380** and samples of a received signal. Based on the discrimination key **380**, the authenticator **850** may analyze the samples to confirm whether they are consistent with the expected signal characteristics of a signal from the trusted transmitter. The authentication process may use similar or the same techniques described above for the DKG **300** to distinguish between samples of signals from the trusted transmitter and other types of samples. However, since the authenticator **350** has access to the discrimination key **380**, it is unnecessary for the authenticator **850** to go through a training process as described above for the DKG **300**. That is, the learned parameters from such training process at the DKG **300** may be included in the discrimination key **380** so that the authenticator **850** may use the same process to authenticate samples from the trusted transmitter without having to be trained.

FIG. **8a** is a block diagram illustrating an exemplary communication device **800** configured to authenticate a signal from a trusted transmitter. The device **800** has a receiver **810** that receives an incoming transmission (e.g., from an antenna). The receiver **810** decodes the incoming transmission to provide decoded data defining the information carried by the received signal. A sampler circuit **320** obtains a copy of the incoming transmission (e.g., either the same antenna for the receiver **810** or from another antenna). The sampler circuit **320** takes a sample of the incoming

signal and sends the sample to the authenticator **850**. The authenticator **850** uses the discrimination key to populate a discrimination algorithm with discrimination data **480** sufficient to determine if the signal data **430** is from the trusted transmitter **120** or another source. The authenticator **850** may then return an output indicative of whether the signal is from the trusted transmitter **120** or some other transmitter.

The authenticator **850** may have previously stored the discrimination key **380**, may have received the discrimination key **380** as part of the transmission from the trusted transmitter, or may have received the discrimination key **380** by querying some other data source.

The discrimination algorithm of the authenticator **850** may be similar to that of the discriminator **350**, except the authenticator **850** would not be required to use any of the learning functions for learning the discrimination data **480** or generate a discrimination key **380**.

In some embodiments, a receiver **810** may be able to export signal data to the authenticator **850**. For example, FIG. **8b** illustrates another exemplary embodiment of the device **800** similar to that described in FIG. **8a** except as indicated hereafter. In this embodiment, there is no separate sampler circuit **320**. The receiver **810** extracts and provides a sample of the received signal to the authenticator **850**. For example, receiver **810** may be a software-defined radio and have already decoded or have access to appropriate signal samples (e.g., IQ data) without using a separate sampler circuit **320**.

In some embodiments, the authenticator **850** may share resources (e.g., one or more processors or memory) with the receiver **810** or other components of the device **800**.

As noted above, the authenticator **850** is configured to set its authentication algorithm with the associated discrimination data **480** to enable the authenticator **850** to determine which signal samples are from the trusted transmitter. Upon receiving a sample of the signal received the receiver **810**, the authenticator **850** determines if the signal sample is consistent with the learned characteristics of transmitter signals from the trusted transmitter. If so, the authenticator **850** returns to the receiver **810** or other component of the device **800** an indication that the received signal is from a trusted source. In such case, the data from the signal may be processed accordingly. Otherwise, the authenticator **850** provides an indication to the receiver **810** or other component of the device **800** that the received signal is not from a trusted source. In such case, the receiver **810** or other component may discard the signal or the data from the signal or perform other actions as may be desired.

FIG. **9** is a block diagram illustrating an exemplary method of authenticating signals using a discriminator key **380**. At step **910**, signal samples **430** are taken from an incoming signal. These signal samples may come from a separate sampler circuit **320** (as shown in FIG. **8a**) or directly from the receiver **810** (as shown in FIG. **8b**). At step **920**, the signal samples are sent to the authenticator **850**. This information can be sent via a serial bus, memory allocation, via an I/O interface, or over a network (e.g., after being captured by methods described in FIGS. **8a** and **8b**). At step **930**, the authenticator **850** uses discrimination data **480** associated with the discrimination key **380** to validate if the signal samples are from a trusted source (e.g., a transmitter used to generate the discrimination key **380**). At step **940**, the authenticator **850** returns an indicator of the result of the authentication.

The signal data discussed above may take many different forms including analog and digital signals that may be represented in different ways known in the art. In some

embodiments, the signal data includes IQ data. In some embodiments, the signal data includes timing data and IQ data. Timing data may come in many forms including timestamps, order or sequencing indicators, time order arrangement of data, and so forth. As an example, each data point may be defined by an I value (indicative of an in-phase component of the signal) and a Q value (indicative of the corresponding quadrature component of the signal), as described above, along with a timestamp indicating a time of the sample from which the associated I value and Q value were derived. Note that the samples used by the discriminator **350** and the samples used by the authenticator **850** may both include timing data, as described above.

Notably, the signal characteristics of a signal transmitted by the trusted transmitter may slightly change over time, such as due to temperature fluctuations in the transmitter. By treating the sample data as time series data indicative of sampling time, patterns in the changes to the sample data due to temperature fluctuations or other runtime conditions become recognizable and may assist to improve the accuracy at which the discrimination key **380** may be used to authenticate a signal from a trusted source.

In experiments, the generative and discriminative models were trained and tested on the collected sample datasets that had 1024 complex I/Q values per timestamp, generating **2048** features. This size offered excellent identification effectiveness. Larger sample sizes do offer greater accuracy but beyond a sample size of 1024 complex values seemed to be a point to diminishing returns. In such embodiments, the unique pattern of variation of the I/Q imbalances for each radio is captured by the machine-learning algorithm and discrimination data **480** as described earlier.

The signal information can be used to train machine-learning systems such as convolutional neural network (CNN), a fully connected deep neural network (DNN), and a recurrent neural network (RNN). Classification may be performed independent of identification, be performed prior to identification, or be performed subsequent to identification as described above. Initial experimental results reveal that the CNN, DNN, and RNN are able to correctly distinguish between the **8** trusted transmitters from the same manufacture with about 81.6%, 94.6% and 97% accuracy respectively. Better accuracy occurs when comparing transmitter's different manufacturers). Time ordered information can be entered in various ways including but not limited to passing previous results or inputs back into future inputs, use of ordered histories, use of carry over data, and so forth.

Some exemplary machine-learning parameters used during testing and these parameters may be set forth as follows for different implementations as exemplary embodiments. These parameters represent a trade of the best possible accuracy under the constraints of the training time include the following. During each training, the maximum epoch was set to 50 with an early stopping condition, such as, if there is no improvement of validation loss for five consecutive epochs, then the training is stopped. The observed results were achieved through multiple runs of training, indicated that each of the models converged within a given range of the maximum number of epochs. The CNN architecture converged between approximately 45-50 epochs, with seven layers, a learning rate of 104. The DNN architecture converged between approximately 35-40 epochs, with five layers, a learning rate of 10-3. The DNN architecture converged between approximately 35-40 epochs, with five layers, a learning rate of 10-3. The RNN-LSTM architecture converged between approximately 30-35 epochs, with six layers, a learning rate of 103. The RNN-

GRU architecture converged between approximately 30-35 epochs, with six layers, a learning rate of 10-3. Each of the above architectures used a batch size of 128 entries. The listed epoch convergence may be used in some embodiments as train count, meaning training simply is performed the specified number of times or as maximum training count for the given architecture. In some embodiments, additional or fewer epochs may be added to the training counts.

The principles outlined herein can be put together in other ways to preform additional functions. For example, a series of signal samples from a collection of different trusted transmitters can be used to train a discriminator **350**. The discriminator **350** may use machine-learning techniques common in machine classification techniques such as those described above to distinguish samples of signals transmitted by trusted transmitters from fake samples based on the same types of signal information described above. Such classification could be used to identify a transmitter as one among a set of trusted transmitters. In some embodiments, such classifications of a set of signal samples a group of trusted transmitters may offer diminished identification accuracy. The issue of diminished accuracy may be resolved by using results of the classification determination to select an associated discrimination key **380** based on the classification results. The selected discrimination key **380** may then be used to authenticate the signal sample's source.

In any event, for some embodiments, rather than training a discriminator **350** to learn the signal characteristics of a signal from a single trusted transmitter, the discriminator **350** may be trained to learn the signal characteristics of a group of trusted transmitters. The same techniques described above may be used for such training except that the discriminator **350** receives actual samples from each transmitter of a group of trusted transmitters rather than a single trusted transmitter. That is, each actual sample from a trusted transmitter of the group of trusted transmitters may be treated as a "good" or trusted sample (i.e., not a fake sample). Thus, if a communication device receives a signal from any of the trusted transmitters, the device's authenticator **850** should authenticate the signal from a trusted source using the same techniques described above.

In some embodiments, a trusted and untrusted classification may be performed first, followed by an identification classification, and then followed an authentication based upon the identification classification.

The foregoing is merely illustrative of the principles of this disclosure and various modifications may be made by those skilled in the art without departing from the scope of this disclosure. The above described embodiments are presented for purposes of illustration and not of limitation. The present disclosure also can take many forms other than those explicitly described herein. For instance, the order of particular steps or the form of particular processes can be changed in some cases to perform equivalent steps. Accordingly, it is emphasized that this disclosure is not limited to the explicitly disclosed methods, systems, and apparatuses, but is intended to include variations to and modifications thereof, which are within the spirit of the following claims.

As a further example, variations of apparatus or process parameters (e.g., dimensions, configurations, components, process step order, etc.) may be made to further optimize the provided structures, devices and methods, as shown and described herein. In any event, the structures and devices, as well as the associated methods, described herein have many applications. Therefore, the disclosed subject matter should not be limited to any single embodiment described herein,

11

but rather should be construed in breadth and scope in accordance with the appended claims.

Now, therefore, the following is claimed:

1. A system for generating discrimination data that is based on hardware characteristics of a transmitter for use in authenticating the transmitter, comprising:

a sampling circuit configured to sample a signal transmitted by the transmitter, thereby defining a plurality of signal samples of the signal, wherein the signal transmitted by the transmitter is modulated with quadrature amplitude modulation (QAM), and wherein each of the plurality of signal samples of the signal comprises a measurement of an in-phase component (I) of the signal and a measurement of a quadrature component of the signal;

a generator configured to provide a plurality of false signal samples; and

a discriminator configured to receive the plurality of signal samples of the signal and the plurality of false signal samples, the discriminator having at least one processor programmed with instructions that, when executed by the at least one processor, cause the discriminator to implement a machine-learning algorithm for learning to discriminate between the plurality of signal samples of the signal and the plurality of false signal samples based on a signal pattern unique to the transmitter represented by in-phase components and quadrature components of signals transmitted by the transmitter, wherein the discriminator is configured to generate the discrimination data for use in authenticating the transmitter based on truth data indicating whether decisions by the discriminator in discriminating between the plurality of signal samples of the signal and the plurality of false signal samples are correct.

2. The system of claim 1, wherein the generator is configured to receive the plurality of signal samples of the signal and to generate the plurality of false signal samples based on the plurality of signal samples of the signal.

3. The system of claim 1, wherein each of the plurality of signal samples of the signal comprises a timestamp.

4. A system for generating discrimination data that is based on hardware characteristics of a transmitter for use in authenticating the transmitter, comprising:

a sampling circuit configured to sample a signal transmitted by the transmitter, thereby defining a first plurality of signal samples of the signal, wherein the signal transmitted by the transmitter is modulated with quadrature amplitude modulation (QAM), and wherein each of the plurality of signal samples of the signal comprises a measurement of an in-phase component (I) of the signal and a measurement of a quadrature component of the signal;

a generator configured to provide a second plurality of signal samples; and

a discriminator configured to receive the first plurality of signal samples and the second plurality of signal samples, the discriminator having at least one processor programmed with instructions that, when executed by the at least one processor, cause the discriminator to implement a machine-learning algorithm for learning to discriminate between the first plurality of signal

12

samples of the signal and the second plurality of signal samples based on a signal pattern unique to the transmitter represented by in-phase components and quadrature components of signals transmitted by the transmitter, wherein the discriminator is configured to generate the discrimination data for use in authenticating the transmitter based on data used in the machine-learning algorithm for discriminating between the first plurality of signal samples and the second plurality of signal samples.

5. The system of claim 4, wherein the generator is configured to generate the second plurality of signal samples, wherein the discriminator is configured to provide feedback to the generator, and wherein at least one of the second plurality of signal samples is based on the feedback from the discriminator.

6. The system of claim 4, further comprising a communication device configured to receive a signal, the communication device having an authenticator configured to receive the discrimination data and a third plurality of signal samples from the received signal, the authenticator configured to analyze the third plurality of signal samples based on the discriminator data for authenticating the received signal.

7. The system of claim 4, wherein the discrimination data further comprises an encoding of the parameters for distinguishing between signals from the transmitter and other signals.

8. The system of claim 4, wherein the generator is configured to receive the first plurality of signal samples and to generate at least one of the second plurality of signal samples based on the first plurality of signal samples.

9. The system of claim 4, wherein each of the first plurality of signal samples comprises a timestamp.

10. A method for generating discrimination data that is based on hardware characteristics of a transmitter for use in authenticating the transmitter, comprising:

sampling a signal transmitted by the transmitter, thereby defining a plurality of signal samples of the signal, wherein the signal transmitted by the transmitter is modulated with quadrature amplitude modulation (QAM), and wherein each of the plurality of signal samples of the signal comprises a measurement of an in-phase component (I) of the signal and a measurement of a quadrature component of the signal;

receiving a plurality of false signal samples;

receiving truth data;

discriminating between the plurality of signal samples of the signal and the false signal samples based on a signal pattern unique to the transmitter represented by in-phase components and quadrature components of signals transmitted by the transmitter;

determining whether decisions in the discriminating are correct based on the truth data; and

generating the discrimination data for use in authenticating the transmitter based on the determining.

11. The method of claim 10, further comprising generating the plurality of false signal samples based on the plurality of signal samples of the signal.

12. The method of claim 10, wherein each of the plurality of signal samples of the signal comprises a timestamp.

* * * * *